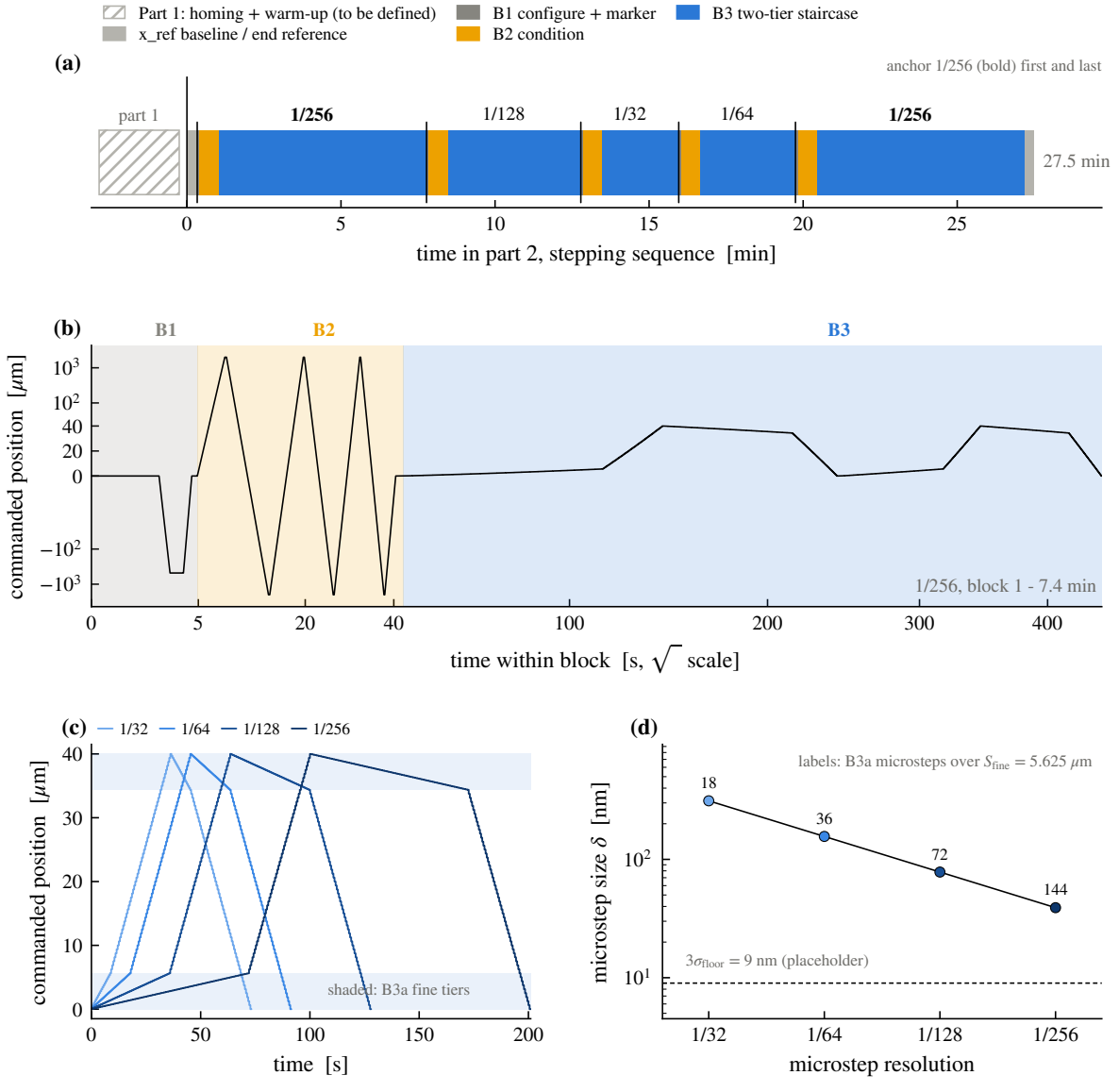
Part 1: homing + warm-up (to be defined)
x_ref baseline / end reference
B1 configure + marker
B2 condition
B3 two-tier staircase
(a)
anchor 1/256 (bold) first and last
part 1
1/256
1/128
1/32
1/64
1/256
27.5 min
0
5
10
15
20
25
time in part 2, stepping sequence [min]
(b)
B1
B2
B3
commanded position [μm]
10^3
10^2
40
20
0
−10^2
−10^3
1/256, block 1 - 7.4 min
0
5
20
40
100
200
300
400
time within block [s, √ scale]
(c)
1/32
1/64
1/128
1/256
commanded position [μm]
40
30
20
10
0
shaded: B3a fine tiers
0
50
100
150
200
time [s]
(d)
labels: B3a microsteps over S_fine = 5.625 μm
18
36
72
144
microstep size δ [nm]
10^2
10^1
3σ_floor = 9 nm (placeholder)
1/32
1/64
1/128
1/256
microstep resolution